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THERMAL-HYDRAULICS IN THE HOT SUB-CHANNEL OF AN AP1000 REACTOR

MATTEO ARMANDO
TOMMASO COGOZZO
FEDERICO PALA

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List of Main Equations

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Introduction

Objective of the Assignment

The objective of this work is to evaluate the steady-state thermal-hydraulic behaviour of the hot subchannel of an AP1000 Pressurized Water Reactor under nominal Beginning of Life (BOL) conditions. The analysis focuses on the most limiting subchannel, where the combined effect of local power peaking and reduced coolant flow produces the highest thermal load.

The calculation is performed using a one-dimensional axial model. Cross-flow between neighbouring subchannels is neglected; therefore, the hot subchannel is treated as hydraulically isolated. This is a conservative simplification, because lateral mixing is not allowed to reduce the local coolant enthalpy rise.

The main quantities evaluated in the analysis are:

- coolant specific enthalpy, temperature and equilibrium quality;
- cladding outer and inner surface temperatures;
- onset of nucleate boiling and bubble detachment position;
- flow quality and void fraction in the boiling region;
- gap conductance, fuel surface temperature and fuel centerline temperature;
- critical heat flux, DNBR profile and minimum DNBR;
- verification of the relevant thermal limits.

The computational procedure follows a progressive thermal-hydraulic sequence. First, the average and maximum volumetric heat generation rates are calculated from the AP1000 nominal power and the assigned heat flux hot-channel factor. Then, the coolant enthalpy and temperature profiles are obtained using the chopped cosine axial power distribution. The cladding outer temperature is determined by comparing the single-phase forced-convection prediction with the Jens-Lottes boiling correlation. Once the onset of nucleate boiling is identified, the Griffith criterion is used to locate the bubble detachment point. Downstream of this location, the flow quality and the void fraction are evaluated.

The solid temperature field is obtained by solving the radial heat transfer problem through the cladding, the gas gap and the fuel pellet. The gap conductance is evaluated by considering gas conduction and thermal radiation, while the fuel centerline temperature is obtained by integrating the temperature-dependent thermal conductivity of uranium dioxide. Finally, the critical heat flux is calculated using the W-3 correlation, including the spacer grid correction and the Tong/Lin correction for non-uniform axial heat flux. The DNBR profile is then obtained by comparing the corrected critical heat flux with the actual local surface heat flux.

The AP1000 Reactor

The AP1000 is a Generation III+ Pressurized Water Reactor developed by Westinghouse. It is based on the standard PWR concept, with a pressurized primary circuit, light water as coolant and moderator, and a core composed of square fuel assemblies.

A distinctive feature of the AP1000 design is the extensive use of passive safety systems. These systems rely on natural driving forces such as gravity, natural circulation and compressed gas, reducing the dependence on active mechanical components and external power supplies during accident conditions.

The AP1000 core considered in this work is composed of 157 fuel assemblies. Each assembly has a 17×17 lattice, with 264 fuel rods and the remaining positions occupied by guide thimbles

and instrumentation tubes. The thermal-hydraulic calculation is therefore based on the simplified representation of an internal hot subchannel associated with one fuel rod. The operational and geometrical parameters used in the model are introduced in Section 1.

1 Input Data and Subchannel Modelling

The thermal-hydraulic analysis of the AP1000 hot subchannel requires the definition of the nominal plant operating conditions, the fuel rod geometry, the subchannel flow area and the conservative hot-channel assumptions. All dimensional quantities originally provided in Imperial units are converted into SI units before being implemented in the MATLAB model.

The calculation is performed on a representative internal hot subchannel associated with one fuel rod. Cross-flow with neighbouring subchannels is neglected, so the model is one-dimensional in the axial direction. This assumption is conservative because lateral mixing is not allowed to reduce the local coolant enthalpy rise.

1.1 Plant Operational Parameters

The AP1000 nominal operating conditions used in the calculation are summarized in Table 1. The total coolant flow rate is corrected by subtracting the bypass fraction, since only the coolant flowing through the active core directly contributes to fuel rod heat removal.

Table 1: AP1000 plant operational parameters used in the hot subchannel calculation.

Parameter	Imperial value	SI value
Core thermal power, Q_{th}	3400 MW _{th}	3400 MW _{th}
Nominal system pressure, P	2250 psia	155.13 bar
Inlet coolant temperature, T_{in}	535.0 °F	279.44 °C
Total coolant flow rate, $\dot{m}_{tot}$	113.5×10^6 lbm/hr	14300.8 kg/s
Core bypass fraction, f_{bypass}	5.9%	0.059
Heat generated in fuel, η_f	97.4%	0.974

The active core mass flow rate is therefore:

$$\dot{m}_{core} = \dot{m}_{tot} (1 - f_{bypass}) \quad (1.1)$$

Using the assigned data:

$$\dot{m}_{core} = 13457 \text{ kg/s} \quad (1.2)$$

The factor $\eta_f = 0.974$ indicates that 97.4% of the total thermal power is generated directly inside the fuel pellets. This distinction is important because the fuel temperature calculation is based on the heat generated in the fuel, while the coolant enthalpy balance must account for the total heat transferred to the coolant.

1.2 Fuel Rod and Subchannel Geometry

The AP1000 core consists of 157 fuel assemblies. Each assembly has a 17×17 lattice, with 264 fuel rods. Therefore, the total number of fuel rods is:

$$N_{rods} = N_{assemblies} N_{rods/assembly} = 157 \times 264 = 41448 \quad (1.3)$$

The geometrical data used in the model are reported in Table 2.

Table 2: Fuel rod and subchannel geometrical data.

Parameter	Imperial value	SI value
Active fuel length, H	168.0 in	4.2672 m
Rod pitch, w	0.496 in	0.012598 m
Cladding outer diameter, D_{co}	0.374 in	0.0094996 m
Cladding thickness, t_{clad}	0.0225 in	0.0005715 m
Cladding inner diameter, D_{ci}	0.329 in	0.0083566 m
Fuel pellet diameter, D_f	0.3225 in	0.0081915 m
Diametral gap	0.0065 in	0.0001651 m

The flow area of the square subchannel is calculated by subtracting the fuel rod cross-sectional area from the square lattice area:

$$A_c = w^2 - \frac{\pi}{4} D_{co}^2 \quad (1.4)$$

The heated perimeter is taken as the outer perimeter of the cladding:

$$P_{heated} = \pi D_{co} \quad (1.5)$$

The equivalent heated diameter used in the heat transfer and CHF correlations is therefore:

$$D_h = \frac{4A_c}{P_{heated}} = \frac{4A_c}{\pi D_{co}} \quad (1.6)$$

In the single-phase heat transfer calculation, the same geometrical quantity is also used as the equivalent hydraulic diameter:

$$D_{eq} = D_h \quad (1.7)$$

For the present geometry, the numerical values are:

$$A_c = 8.7843 \times 10^{-5} \text{ m}^2 \quad (1.8)$$

$$D_h = 1.1774 \times 10^{-2} \text{ m} \quad (1.9)$$

This value of D_h is slightly above the nominal validity range of some CHF correlations. It is nevertheless retained as the representative equivalent heated diameter of the AP1000 hot subchannel. The CHF and DNBR results are therefore interpreted as estimates obtained within the empirical limits of the adopted correlation.

The fuel pellet cross-sectional area is:

$$A_f = \frac{\pi}{4} D_f^2 \quad (1.10)$$

1.3 Hot-Channel Power and Flow Assumptions

The average volumetric heat generation rate in the fuel is calculated using only the fraction of the total thermal power generated inside the fuel pellets:

$$q_{v,avg} = \frac{Q_{th}\eta_f}{N_{rods}A_fH} \quad (1.11)$$

The local maximum volumetric heat generation rate is obtained by applying the heat flux hot-channel factor:

$$q_{v,max} = F_Q q_{v,avg} \quad (1.12)$$

where:

$$F_Q = 2.60 \quad (1.13)$$

The resulting values are:

$$q_{v,avg} = 3.553 \times 10^8 \text{ W/m}^3 \quad (1.14)$$

$$q_{v,max} = 9.237 \times 10^8 \text{ W/m}^3 \quad (1.15)$$

The average mass flow rate per subchannel is:

$$\dot{m}_{sub,avg} = \frac{\dot{m}_{core}}{N_{rods}} \quad (1.16)$$

which gives:

$$\dot{m}_{sub,avg} = 0.324672 \text{ kg/s} \quad (1.17)$$

A conservative inlet flow maldistribution penalty of 5% is applied to the hot subchannel. Therefore, the hot-channel mass flow rate is:

$$W_{HC} = \dot{m}_{sub,avg}(1 - 0.05) \quad (1.18)$$

$$W_{HC} = 0.308438 \text{ kg/s} \quad (1.19)$$

The corresponding average and hot-channel mass velocities are:

$$G_{avg} = \frac{\dot{m}_{sub,avg}}{A_c} \quad (1.20)$$

$$G_{HC} = \frac{W_{HC}}{A_c} \quad (1.21)$$

with numerical values:

$$G_{avg} = 3696.0 \text{ kg/m}^2\text{s} \quad (1.22)$$

$$G_{HC} = 3511.2 \text{ kg/m}^2\text{s} \quad (1.23)$$

The reduced hot-channel mass velocity is used in the convective heat transfer and CHF calculations. This is conservative because a lower mass velocity decreases the convective heat transfer coefficient and reduces the margin to critical heat flux.

2 Heat Generation and Coolant Thermodynamics

The thermal analysis of the hot subchannel starts from the definition of the axial heat generation profile inside the fuel pellet. The heat generated in the fuel is conducted radially through the pellet, the gas gap and the cladding, and is finally transferred to the coolant flowing upward along the active core.

The axial coordinate is centred at the core mid-plane:

$$-\frac{H}{2} \leq z \leq \frac{H}{2} \quad (2.1)$$

where $z = -H/2$ corresponds to the core inlet and $z = H/2$ to the core outlet.

2.1 Axial Power Profile

The axial distribution of volumetric heat generation is represented by a chopped cosine profile. This shape accounts for the fact that the neutron flux does not become exactly zero at the physical boundaries of the active fuel, because of neutron leakage and reflection effects.

The extrapolated height is calculated as:

$$H_e = H + 1.42\lambda_{tr} + 2\delta_{ref} \quad (2.2)$$

where λ_{tr} is the neutron transport mean free path and δ_{ref} is the reflector extrapolation distance.

The local volumetric heat generation rate in the hot subchannel is therefore:

$$q_v(z) = q_{v,max} \cos\left(\frac{\pi z}{H_e}\right) \quad (2.3)$$

where $q_{v,max}$ is the maximum local volumetric heat generation rate defined in Section 1.3.

Since $q_v(z)$ is defined as the volumetric heat generation rate inside the fuel, the local linear heat rate generated in the fuel is:

$$q'_f(z) = q_v(z)A_f \quad (2.4)$$

The corresponding cladding outer surface heat flux is obtained by dividing the local linear heat rate by the heated perimeter of the fuel rod:

$$q''(z) = \frac{q'_f(z)}{\pi D_{co}} = q_v(z) \frac{A_f}{\pi D_{co}} \quad (2.5)$$

In the present calculation, the maximum surface heat flux is:

$$q''_{max} = 1.631 \text{ MW/m}^2 \quad (2.6)$$

and it occurs approximately at the core mid-plane, where the chopped cosine profile reaches its maximum value.

2.2 Coolant Specific Enthalpy

The coolant specific enthalpy $i(z)$ increases along the axial coordinate because of the heat transferred from the fuel rod to the coolant. The hot-channel mass flow rate W_{HC} , defined in Section 1.3, is used in the energy balance.

The average volumetric heat generation rate was calculated using the fraction of power generated directly in the fuel, $\eta_f = 0.974$. Therefore, when evaluating the coolant enthalpy rise, the heat deposited outside the fuel must also be included. This is done by applying the correction factor $1/\eta_f$ to the fuel-generated linear heat rate.

The differential energy balance for the coolant is:

$$\frac{di}{dz} = \frac{1}{\eta_f} \frac{q_v(z)A_f}{W_{HC}} \quad (2.7)$$

Substituting the chopped cosine profile and integrating from the inlet of the active core, $z = -H/2$, to a generic axial position z , gives:

$$i(z) = i_{in} + \frac{1}{\eta_f} \frac{q_{v,max}A_f}{W_{HC}} \int_{-H/2}^z \cos\left(\frac{\pi z'}{H_e}\right) dz' \quad (2.8)$$

The analytical integration gives the expression implemented in the MATLAB code:

$$i(z) = i_{in} + \frac{1}{\eta_f} \frac{q_{v,max} A_f H_e}{W_{HC} \pi} \left[\sin\left(\frac{\pi z}{H_e}\right) + \sin\left(\frac{\pi H}{2H_e}\right) \right] \quad (2.9)$$

The inlet specific enthalpy i_{in} is evaluated at the nominal pressure and inlet temperature using the XSteam thermodynamic property library.

2.3 Coolant Temperature and Equilibrium Quality

Once the enthalpy profile is known, the bulk coolant temperature is obtained from the thermodynamic relation:

$$T_{coolant}(z) = T(P, i(z)) \quad (2.10)$$

The coolant temperature is limited by the saturation temperature corresponding to the nominal system pressure. Therefore, the implemented temperature profile is:

$$T_{coolant}(z) = \min [T(P, i(z)), T_{sat}] \quad (2.11)$$

At the nominal pressure of 155.13 bar, the saturation temperature is:

$$T_{sat} = 344.86 \text{ }^\circ\text{C} \quad (2.12)$$

The equilibrium quality is calculated from the local coolant enthalpy as:

$$x_{eq}(z) = \frac{i(z) - i_{ls}}{i_{vs} - i_{ls}} = \frac{i(z) - i_{ls}}{H_{fg}} \quad (2.13)$$

where i_{ls} is the saturated liquid enthalpy, i_{vs} is the saturated vapour enthalpy and H_{fg} is the latent heat of vaporization at the nominal pressure.

Negative values of x_{eq} indicate subcooled liquid conditions. Positive values indicate that, under thermodynamic equilibrium, a vapour fraction would be present. In the final calculation, the coolant reaches saturation in the upper part of the active core and the outlet equilibrium quality is:

$$x_{eq,out} = 0.04512 \quad (2.14)$$

The corresponding outlet coolant temperature is equal to the saturation temperature:

$$T_{coolant,out} = 344.86 \text{ }^\circ\text{C} \quad (2.15)$$

3 Cladding Outer Temperature and Boiling Regimes

The cladding outer temperature is a key quantity in the hot subchannel analysis, because it determines the heat transfer regime between the fuel rod and the coolant. In the lower part of the core, heat transfer occurs by forced single-phase convection. When the wall temperature becomes sufficiently high, subcooled nucleate boiling starts at the cladding surface and the heat transfer mechanism changes.

The outer cladding temperature is therefore evaluated by comparing two limiting models:

- the single-phase forced-convection wall temperature, $T_{co,SP}$;
- the wall temperature predicted by the Jens–Lottes nucleate boiling correlation, $T_{co,JL}$.

The actual cladding outer temperature is then taken as the lower of these two values.

3.1 Single-Phase Heat Transfer

Before the onset of nucleate boiling, the heat generated in the fuel rod is removed by forced convection to the liquid coolant. The local surface heat flux at the cladding outer surface has already been defined in Section 2.1 as:

$$q''(z) = q_v(z) \frac{A_f}{\pi D_{co}} \quad (3.1)$$

The single-phase heat transfer coefficient is evaluated using a modified Dittus–Boelter correlation for rod bundles. A geometry-dependent coefficient is introduced to account for the square lattice arrangement:

$$C_{array} = 0.042 \left(\frac{w}{D_{co}} \right) - 0.024 \quad (3.2)$$

The Reynolds number is calculated using the conservative hot-channel mass velocity:

$$Re(z) = \frac{G_{HC} D_{eq}}{\mu(z)} \quad (3.3)$$

where D_{eq} is the equivalent hydraulic diameter, $\mu(z)$ is the dynamic viscosity of the liquid coolant and G_{HC} is the hot-channel mass velocity.

The Prandtl number is:

$$Pr(z) = \frac{c_p(z) \mu(z)}{k_{water}(z)} \quad (3.4)$$

The Nusselt number is then obtained as:

$$Nu(z) = C_{array} Re(z)^{0.8} Pr(z)^{0.4} \quad (3.5)$$

and the corresponding single-phase heat transfer coefficient is:

$$h_{SP}(z) = \frac{Nu(z) k_{water}(z)}{D_{eq}} \quad (3.6)$$

The single-phase prediction of the cladding outer temperature is therefore:

$$T_{co,SP}(z) = T_{coolant}(z) + \frac{q''(z)}{h_{SP}(z)} \quad (3.7)$$

This expression is valid only as long as the heat transfer remains in the single-phase liquid regime.

3.2 Jens–Lottes Correlation

When the cladding surface becomes sufficiently hot, vapour bubbles start to nucleate at the wall even if the bulk coolant is still subcooled. This condition is called subcooled nucleate boiling.

The wall temperature required to sustain nucleate boiling is evaluated using the Jens–Lottes correlation:

$$T_{co,JL}(z) = T_{sat} + 25 \left(\frac{q''(z)}{10^6} \right)^{0.25} \exp \left(-\frac{P}{62} \right) \quad (3.8)$$

where:

- T_{sat} is the saturation temperature at the nominal system pressure, expressed in °C;
- $q''(z)$ is the local surface heat flux in W/m², divided by 10⁶ to obtain MW/m²;

- P is the system pressure expressed in bar.

The Jens–Lottes correlation gives the wall temperature in the nucleate boiling regime. In this regime, the wall temperature depends mainly on local heat flux and pressure, and becomes much less sensitive to the bulk coolant temperature than in single-phase convection.

3.3 Onset of Nucleate Boiling

The onset of nucleate boiling is identified by comparing the single-phase wall temperature with the Jens–Lottes wall temperature. The ONB location is the first axial point where the single-phase prediction exceeds the nucleate boiling prediction:

$$T_{co,SP}(z) > T_{co,JL}(z) \quad (3.9)$$

In the MATLAB implementation, this condition is detected as the first axial node satisfying:

$$z = z_{ONB} \quad (3.10)$$

The actual cladding outer temperature is then calculated as:

$$T_{co}(z) = \min [T_{co,SP}(z), T_{co,JL}(z)] \quad (3.11)$$

Therefore:

- for $z < z_{ONB}$, heat transfer is single-phase and $T_{co} = T_{co,SP}$;
- for $z \geq z_{ONB}$, nucleate boiling controls the wall temperature and $T_{co} = T_{co,JL}$.

In the final calculation, the onset of nucleate boiling is found at:

$$z_{ONB} = -0.184 \text{ m} \quad (3.12)$$

This point is important because it marks the beginning of wall vapour generation and represents the lower integration boundary for the Tong/Lin non-uniform heat flux correction used later in the CHF calculation.

4 Flow Quality and Void Fraction Evaluation

After the onset of nucleate boiling, vapour bubbles start to form at the cladding surface. However, wall boiling does not immediately imply that vapour is transported in the bulk coolant flow. In the first part of the subcooled boiling region, bubbles remain attached to the heated wall and the actual flow quality is still zero.

For this reason, the two-phase calculation is divided into two steps. First, the bubble detachment point is identified using the Griffith criterion. Then, downstream of this point, the actual flow quality and the void fraction are evaluated.

4.1 Bubble Detachment Point

The bubble detachment point z_D marks the beginning of macroscopic vapour transport in the bulk coolant. Upstream of this point, vapour bubbles are present at the wall, but they do not contribute to the flow quality.

The detachment condition is evaluated using the Griffith model. According to this model, bubbles detach from the heated surface when the bulk liquid temperature reaches the detachment liquid temperature:

$$T_{l,D}(z) = T_{sat} - \frac{q''(z)}{5h_{SP}(z)} \quad (4.1)$$

where $q''(z)$ is the local cladding surface heat flux and $h_{SP}(z)$ is the single-phase heat transfer coefficient.

The detachment point is found as the first axial location, after ONB, where:

$$T_{coolant}(z_D) > T_{l,D}(z_D) \quad (4.2)$$

Therefore, for $z < z_D$, bubbles are attached to the wall and the flow quality is zero; for $z \geq z_D$, detached bubbles are entrained into the bulk flow and the flow quality starts increasing.

In the final calculation, the bubble detachment point is located at:

$$z_D = 0.937 \text{ m} \quad (4.3)$$

4.2 Flow Quality Calculation

The equilibrium quality x_{eq} is a thermodynamic quantity based only on the local coolant enthalpy. In subcooled boiling, however, the actual vapour mass fraction transported by the flow is lower than the equilibrium value, because part of the heat is still used to heat the subcooled liquid and part of the vapour may condense in the liquid bulk.

The actual flow quality is therefore set to zero before bubble detachment:

$$x(z) = 0 \quad \text{for } z < z_D \quad (4.4)$$

Downstream of detachment, the wall heat flux is divided into three contributions:

$$q''(z) = q_e''(z) + q_a''(z) + q_{sp}''(z) \quad (4.5)$$

where q_e'' is the evaporation heat flux, q_a'' is the agitation heat flux associated with bubble motion, and q_{sp}'' is the single-phase sensible heat flux transferred to the subcooled liquid.

The single-phase sensible heat flux is calculated as:

$$q_{sp}''(z) = h_{SP}(z) \max[0, T_{sat} - T_{coolant}(z)] \quad (4.6)$$

The Bowring parameter is defined as:

$$\epsilon = \frac{q_a''}{q_e''} \quad (4.7)$$

For high-pressure PWR conditions, the Bowring value used in the calculation is:

$$\epsilon_{Bowring} = 1.6 \quad (4.8)$$

Therefore, the evaporation heat flux is obtained as:

$$q_e''(z) = \frac{q''(z) - q_{sp}''(z)}{1 + \epsilon_{Bowring}} \quad (4.9)$$

In the MATLAB implementation, negative values of q_e'' are not allowed, since they would correspond to no net vapour production.

The flow quality is then obtained by integrating the evaporation contribution from the detachment point to the generic axial position z :

$$x(z) = \int_{z_D}^z \frac{q_e''(z') P_{heated}}{W_{HC} H_{fg}} dz' \quad (4.10)$$

where $P_{heated} = \pi D_{co}$, W_{HC} is the hot-channel mass flow rate and H_{fg} is the latent heat of vaporization at the nominal pressure.

In the final calculation, the maximum flow quality is reached at the core outlet:

$$x_{max} = 0.03015 \quad (4.11)$$

4.3 Bowring–Rouhani Epsilon Comparison

Although the final flow quality calculation is performed using the Bowring constant value $\epsilon_{Bowring} = 1.6$, the Rouhani formulation is also evaluated as a comparison for the heat partitioning coefficient.

The Rouhani coefficient is calculated from the local thermodynamic state as:

$$\epsilon_{Rouhani}(z) = \frac{\rho_l(z)}{\rho_g H_{fg}} [i_{ls} - i(z)] \quad (4.12)$$

When the coolant approaches saturation, the enthalpy difference $i_{ls} - i(z)$ tends to zero. If the local enthalpy exceeds the saturated liquid enthalpy, the Rouhani expression would become negative. Since a negative heat partitioning coefficient is not physically meaningful in this model, the implemented value is limited as:

$$\epsilon_{Rouhani}(z) = \max \left[0, \frac{\rho_l(z)}{\rho_g H_{fg}} (i_{ls} - i(z)) \right] \quad (4.13)$$

This comparison is useful to show how the heat partitioning evolves as the coolant approaches saturated conditions. However, the final reported flow quality is obtained using the Bowring constant value, while the Rouhani formulation is retained only as a sensitivity comparison.

4.4 Void Fraction Analysis

The void fraction α is the volumetric fraction of the subchannel occupied by vapour. It is not equal to the flow quality, because the vapour and liquid phases have very different densities and may move with different velocities.

Between ONB and bubble detachment, vapour bubbles are present at the heated wall but are not yet transported in the bulk flow. In this region, the flow quality is zero, but a small wall void fraction is still present. The bubble detachment radius is estimated as:

$$R_d = \frac{2.37 \times 10^{-3}}{P^{0.237}} \quad (4.14)$$

with P expressed in bar. The wall bubble layer thickness is:

$$\delta_{layer} = 0.0666 R_d \quad (4.15)$$

and the void fraction at detachment is:

$$\alpha_{det} = \frac{4\delta_{layer}}{D_{eq}} \quad (4.16)$$

Between z_{ONB} and z_D , the void fraction is assumed to increase linearly from zero to α_{det} :

$$\alpha(z) = \alpha_{det} \frac{z - z_{ONB}}{z_D - z_{ONB}} \quad z_{ONB} \leq z < z_D \quad (4.17)$$

For $z \geq z_D$, detached bubbles are transported by the coolant and the void fraction is calculated from the actual flow quality. The first model is the Zuber–Findlay drift-flux model:

$$\alpha_{ZF}(z) = \frac{x(z)/\rho_g}{C_0 [x(z)/\rho_g + (1 - x(z))/\rho_l] + V_{gj}/G_{HC}} \quad (4.18)$$

where $C_0 = 1.13$ and:

$$V_{gj} = 1.18 \left[\frac{\sigma g(\rho_l - \rho_g)}{\rho_l^2} \right]^{1/4} \quad (4.19)$$

The second model is the slip-ratio model with the Fauske slip ratio:

$$S = \sqrt{\frac{\rho_l}{\rho_g}} \quad (4.20)$$

The corresponding void fraction is:

$$\alpha_{Slip}(z) = \frac{x(z)}{x(z) + (1 - x(z)) \frac{\rho_g}{\rho_l} S} \quad (4.21)$$

To avoid an artificial discontinuity at the detachment point, the calculated bulk void fraction downstream of detachment is not allowed to fall below the wall void fraction already present at detachment:

$$\alpha(z) = \max [\alpha_{det}, \alpha_{calc}(z)] \quad (4.22)$$

In the final calculation, the maximum void fractions are reached at the core outlet:

$$\alpha_{ZF,max} = 0.13377 \quad (4.23)$$

$$\alpha_{Slip,max} = 0.06978 \quad (4.24)$$

The void fraction increases more rapidly than the flow quality because the vapour density is much lower than the liquid density. Consequently, even a small vapour mass fraction occupies a significant volume fraction. The Zuber–Findlay model predicts a larger void fraction than the slip-ratio model because it includes drift-flux and phase-distribution effects, while the Fauske slip-ratio model imposes a stronger relative motion between vapour and liquid.

5 Iterative Calculation of Internal Temperatures

After the coolant state and the cladding outer temperature have been determined, the calculation proceeds radially inward through the fuel rod. The objective is to evaluate the cladding inner temperature, the gap conductance, the fuel surface temperature and the fuel centerline temperature at each axial node.

The radial temperature calculation is coupled with the thermo-mechanical behaviour of the fuel rod. The effective fuel-cladding gap depends on the thermal expansion of the fuel pellet, the thermal expansion of the cladding and the elastic deformation of the cladding due to the pressure difference between the primary coolant and the internal helium pressure. Since these quantities depend on the unknown solid temperatures, an iterative procedure is required.

5.1 Inner Cladding Temperature

The inner cladding temperature T_{ci} is obtained by solving the steady-state radial conduction equation through the cylindrical cladding wall. Heat generation inside the cladding is neglected, so the heat conducted through the cladding is equal to the local heat generated in the fuel pellet.

$$\int_{T_{co}(z)}^{T_{ci}(z)} k_{cl}(T) dT = \frac{q'_f(z)}{2\pi} \ln \left(\frac{D_{co}}{D_{ci}} \right) \quad (5.1)$$

The Zircaloy thermal conductivity is assumed to vary linearly with temperature:

$$k_{cl}(T) = 11.45 + 0.01425T \quad (5.2)$$

Substituting Eq. 5.2 into Eq. 5.1 gives a quadratic equation in T_{ci} :

$$\frac{0.01425}{2} T_{ci}^2 + 11.45 T_{ci} - \left[\frac{0.01425}{2} T_{co}^2 + 11.45 T_{co} + \frac{q'_f(z)}{2\pi} \ln \left(\frac{D_{co}}{D_{ci}} \right) \right] = 0 \quad (5.3)$$

The physically meaningful positive root is selected at each axial node. In the final calculation:

$$T_{ci,max} = 406.23 \text{ } ^\circ\text{C} \quad (5.4)$$

5.2 Thermo-Mechanical Deformation of Fuel and Cladding

The diametral thermal expansion of the fuel pellet is calculated as:

$$\Delta D_f = D_f \alpha_f(T_{f,avg})(T_{f,avg} - T_a) \quad (5.5)$$

with:

$$\alpha_f(T_{f,avg}) = 7.87 \times 10^{-6} + 3.9 \times 10^{-9} T_{f,avg} \quad (5.6)$$

The diametral thermal expansion of the cladding inner surface is:

$$\Delta D_{ci,th} = D_{ci} \alpha_{cl}(T_{cl,avg})(T_{cl,avg} - T_a) \quad (5.7)$$

with:

$$\alpha_{cl}(T_{cl,avg}) = 5.62 \times 10^{-6} + 3.162 \times 10^{-9} T_{cl,avg} \quad (5.8)$$

The elastic deformation of the cladding is evaluated using thick-cylinder theory. The diameter ratio is:

$$\gamma = \frac{D_{co}}{D_{ci}} \quad (5.9)$$

The Young modulus of Zircaloy is:

$$E_{Zr} = 1.148 \times 10^{11} - 5.99 \times 10^7 T_{cl,avg,K} \quad (5.10)$$

The elastic strain at the inner cladding surface is:

$$\varepsilon_{el} = \frac{1}{E_{Zr}} \frac{1}{\gamma^2 - 1} \left[p_i \left((1 - \nu_{Zr}) + (1 + \nu_{Zr}) \gamma^2 \right) - 2\gamma^2 p_e \right] \quad (5.11)$$

The corresponding elastic diametral variation is:

$$\Delta D_{ci,el} = D_{ci} \varepsilon_{el} \quad (5.12)$$

The hot gap thickness is finally calculated as:

$$gap_{hot} = \frac{(D_{ci} + \Delta D_{ci,th} + \Delta D_{ci,el}) - (D_f + \Delta D_f)}{2} \quad (5.13)$$

If the computed gap becomes negative, the code sets $gap_{hot} = 0$, representing the limiting condition of fuel-cladding contact.

5.3 Gap Conductance

The heat transfer across the fuel-cladding gap is modelled by combining helium gas conduction and thermal radiation:

$$h_{gap,total} = h_{gap,cond} + h_{rad} \quad (5.14)$$

The helium thermal conductivity is:

$$k_{He} = 0.1763 \times 10^{-2} T_{gap,avg,K}^{0.77163} \quad (5.15)$$

The conductive gap heat transfer coefficient is calculated using a Ross-Stoute type formulation:

$$h_{gap,cond} = \frac{k_{He}}{gap_{hot} + 2.54 \times 10^{-5}} \quad (5.16)$$

If $gap_{hot} = 0$, the MATLAB code assigns $h_{gap,cond} = 5.0 \times 10^4$ W/m²K to avoid a mathematical singularity and represent strong thermal contact.

The radiative contribution is:

$$h_{rad} = \frac{\sigma_{SB} (T_{fs,K}^4 - T_{ci,K}^4)}{(T_{fs,K} - T_{ci,K}) (1/\varepsilon_f + 1/\varepsilon_{cl} - 1)} \quad (5.17)$$

In the present calculation, unit emissivities are assumed: $\varepsilon_f = \varepsilon_{cl} = 1$.

5.4 Fuel Surface and Centerline Temperature

The heat flux at the fuel pellet surface is:

$$q_f''(z) = \frac{q_f'(z)}{\pi D_f} = q_v(z) \frac{A_f}{\pi D_f} \quad (5.18)$$

Once the total gap conductance is known, the fuel surface temperature is updated from:

$$T_{fs}(z) = T_{ci}(z) + \frac{q_f''(z)}{h_{gap,total}(z)} \quad (5.19)$$

In the final calculation, $T_{fs,max} = 644.11$ °C.

The fuel centerline temperature is obtained by integrating the radial conduction equation in a solid cylinder with internal heat generation:

$$\int_{T_{fs}(z)}^{T_{fcl}(z)} k_{UO_2}(T) dT = \frac{q_f'(z) f}{4\pi} \quad (5.20)$$

The uranium dioxide thermal conductivity is:

$$k_{UO_2}(T) = 100 \left[\frac{1}{11.8 + 0.0238T} + 8.775 \times 10^{-13} T^3 \right] \quad (5.21)$$

The primitive of $k_{UO_2}(T)$ is:

$$K(T) = \frac{100}{0.0238} \ln(11.8 + 0.0238T) + \frac{100 \cdot 8.775 \times 10^{-13}}{4} T^4 \quad (5.22)$$

Therefore, the fuel centerline temperature is found by solving:

$$K(T_{fcl}) = K(T_{fs}) + \frac{q'_f(z)f}{4\pi} \quad (5.23)$$

The nonlinear equation is solved with a Newton–Raphson iteration:

$$T_{fcl}^{(n+1)} = T_{fcl}^{(n)} - \frac{K(T_{fcl}^{(n)}) - K_{target}}{k_{UO_2}(T_{fcl}^{(n)})} \quad (5.24)$$

In the final calculation:

$$T_{fcl,max} = 2094.11 \text{ } ^\circ\text{C} < 2804.44 \text{ } ^\circ\text{C} \quad (5.25)$$

Therefore, the fuel centerline temperature satisfies the thermal limit.

6 Critical Heat Flux and DNBR Analysis

The final part of the thermal-hydraulic analysis consists in evaluating the Critical Heat Flux (CHF) and the Departure from Nucleate Boiling Ratio (DNBR). The local DNBR is:

$$DNBR(z) = \frac{q''_{CHF}(z)}{q''(z)} \quad (6.1)$$

6.1 W-3 Correlation

The critical heat flux is first evaluated using the W-3 correlation. The pressure is used in MPa:

$$p = \frac{P_{bar}}{10} \quad (6.2)$$

The equivalent diameter used in the W-3 correlation is the equivalent heated diameter:

$$D_h = \frac{4A_c}{P_{heated}} = \frac{4A_c}{\pi D_{co}} \quad (6.3)$$

For the present AP1000 subchannel geometry, $D_h = 1.1774 \times 10^{-2}$ m. This value is slightly above the nominal validity range of the W-3 correlation. The W-3 model is retained as the reference CHF correlation for the present PWR hot-subchannel calculation, and the results are interpreted within the empirical validity limits of the model.

The W-3 critical heat flux is calculated as:

$$q''_{c,EU} = T_1 T_2 T_3 T_4 T_5 \quad (6.4)$$

where:

$$T_1 = (2.022 - 0.06238p) + (0.1722 - 0.01427p) \exp[(18.177 - 0.5987p)x_{eq}] \quad (6.5)$$

$$T_2 = (0.1484 - 1.596x_{eq} + 0.1729x_{eq}|x_{eq}|)2.326G_{HC} + 3271 \quad (6.6)$$

$$T_3 = 1.157 - 0.869x_{eq} \quad (6.7)$$

$$T_4 = 0.2664 + 0.8357 \exp(-124.1D_h) \quad (6.8)$$

$$T_5 = 0.8285 + 0.0003413(i_{ls} - i_{in}) \quad (6.9)$$

The result is converted from kW/m² to W/m².

6.2 Spacer Grid and Bundle Corrections

The W-3 value is corrected to account for spacer grid effects:

$$F_S = \left(\frac{P_{psia}}{225.896} \right)^{1/2} (1.445 - 0.0371L) \left[\exp \left((x_{eq} + 0.2)^2 \right) - 0.73 \right] + K_s \left(\frac{G_{imp}}{10^6} \right) \left(\frac{\alpha}{0.019} \right)^{0.35} \quad (6.10)$$

The corrected CHF is:

$$q''_{c,grid}(z) = q''_{c,EU}(z) F_S(z) \quad (6.11)$$

A further correction factor of 0.88 is applied for the 17×17 bundle configuration:

$$q''_{c,grid,17 \times 17}(z) = 0.88 q''_{c,grid}(z) \quad (6.12)$$

6.3 Tong/Lin Correction for Non-Uniform Heat Flux

The actual axial heat flux is non-uniform, so the local CHF is corrected for upstream heat flux history using the Tong/Lin correction:

$$q''_{c,NU}(z) = \frac{q''_{c,grid,17 \times 17}(z)}{F(z)} \quad (6.13)$$

The revised Lin formulation is:

$$C = 0.15 \frac{(1 - x_{eq})^{4.31}}{G^{0.478}} \quad (6.14)$$

where G is expressed in Mlb/hr ft^2 and C is obtained in in^{-1} . In MATLAB, C is converted to m^{-1} .

The factor $F(z)$ is evaluated by numerical integration from the ONB location:

$$F(z) = \frac{C \int_{z_{ONB}}^z q''(z') \exp[-C(z - z')] dz'}{q''(z)[1 - \exp(-CL_c)]} \quad (6.15)$$

The Tong/Lin factor is used only as an internal correction for the CHF. Its standalone plot is not reported, while the CHF comparison plot is shown starting from the ONB location.

6.4 DNBR Profile and Minimum DNBR

The final corrected CHF is:

$$q''_{CHF}(z) = q''_{c,NU}(z) \quad (6.16)$$

The DNBR profile is calculated as:

$$DNBR(z) = \frac{q''_{c,NU}(z)}{q''(z)} \quad (6.17)$$

The minimum DNBR is:

$$MDNBR = 1.967 \quad (6.18)$$

located at:

$$z_{MDNBR} = 1.037 \text{ m} \quad (6.19)$$

The minimum is above the core mid-plane because coolant enthalpy and equilibrium quality increase along the channel, reducing the CHF margin in the upper part of the core even though the actual heat flux decreases after the mid-plane.

6.5 Verification of Thermal Limits

The calculated values are compared with the reference limits:

$$DNBR_{limit,nominal} = 1.85 \quad (6.20)$$

$$DNBR_{limit,overpower} = 1.30 \quad (6.21)$$

The minimum DNBR satisfies both limits:

$$1.967 > 1.85 \quad (6.22)$$

$$1.967 > 1.30 \quad (6.23)$$

The fuel centerline temperature limit is also satisfied:

$$T_{fcl,max} = 2094.11 \text{ }^{\circ}\text{C} < 2804.44 \text{ }^{\circ}\text{C} \quad (6.24)$$

7 Results and Discussion

The thermal-hydraulic model described in the previous sections was implemented in MATLAB and applied to the AP1000 hot subchannel under nominal Beginning of Life conditions. The results include the coolant thermodynamic state, the boiling regime transition, the solid temperature profiles, the flow quality and void fraction, and the final CHF/DNBR verification.

7.1 Axial Temperature Profiles

Figure 1 shows the axial temperature profiles of the coolant, the cladding outer surface, the cladding inner surface, the fuel pellet outer surface and the fuel centerline.

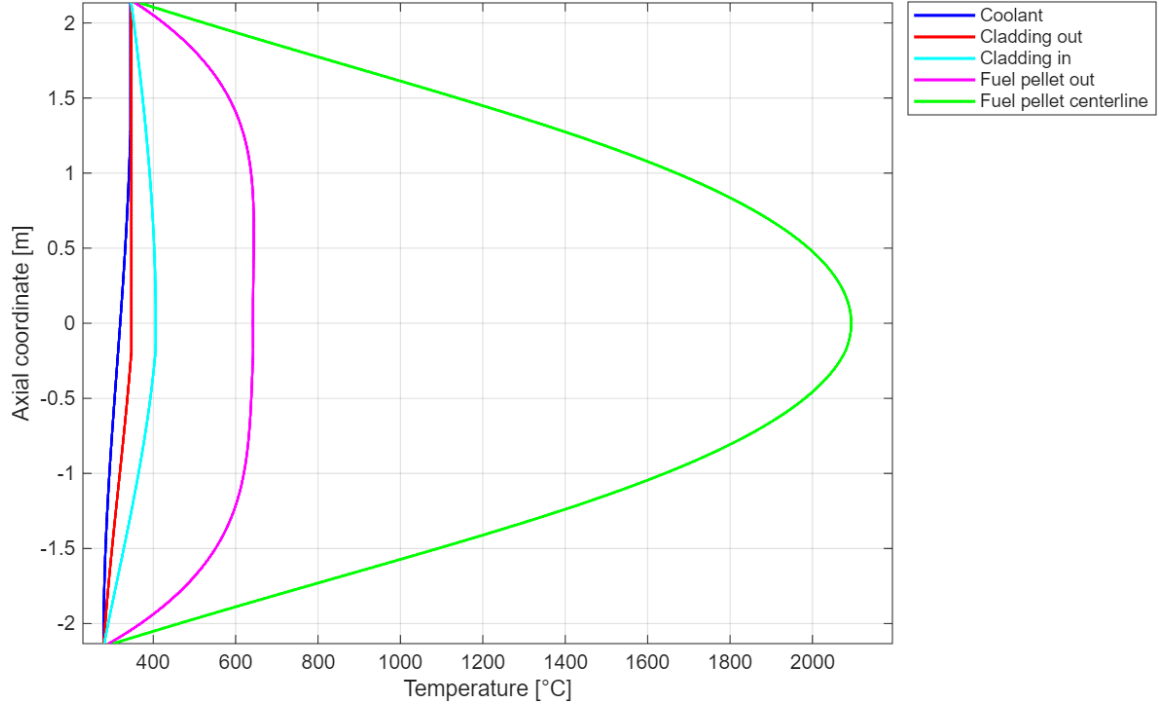


Figure 1: Axial temperature profiles of coolant, cladding and fuel pellet along the AP1000 hot subchannel.

The coolant temperature increases from the inlet value of 279.44 °C and reaches saturation in the upper part of the core. The outlet equilibrium quality is $x_{eq,out} = 0.04512$. The maximum fuel centerline temperature is 2094.11 °C, below the reference uranium dioxide melting temperature of 2804.44 °C.

The different axial trends are physically consistent with the one-dimensional energy balance. The coolant temperature increases monotonically because the enthalpy rise is cumulative along the channel. By contrast, the fuel and cladding temperatures are mainly driven by the local heat generation rate and therefore follow the chopped cosine power profile. The highest fuel centerline temperature occurs close to the axial power peak, where the local heat generation is maximum. A slight downstream shift in some solid-temperature profiles is expected because the coolant entering the upper half of the core is already hotter, reducing the local temperature difference available for heat removal.

7.2 Cladding Outer Temperature and ONB

Figure 2 compares the cladding outer temperature predicted by single-phase convection, the Jens-Lottes boiling correlation and the final selected cladding outer temperature.

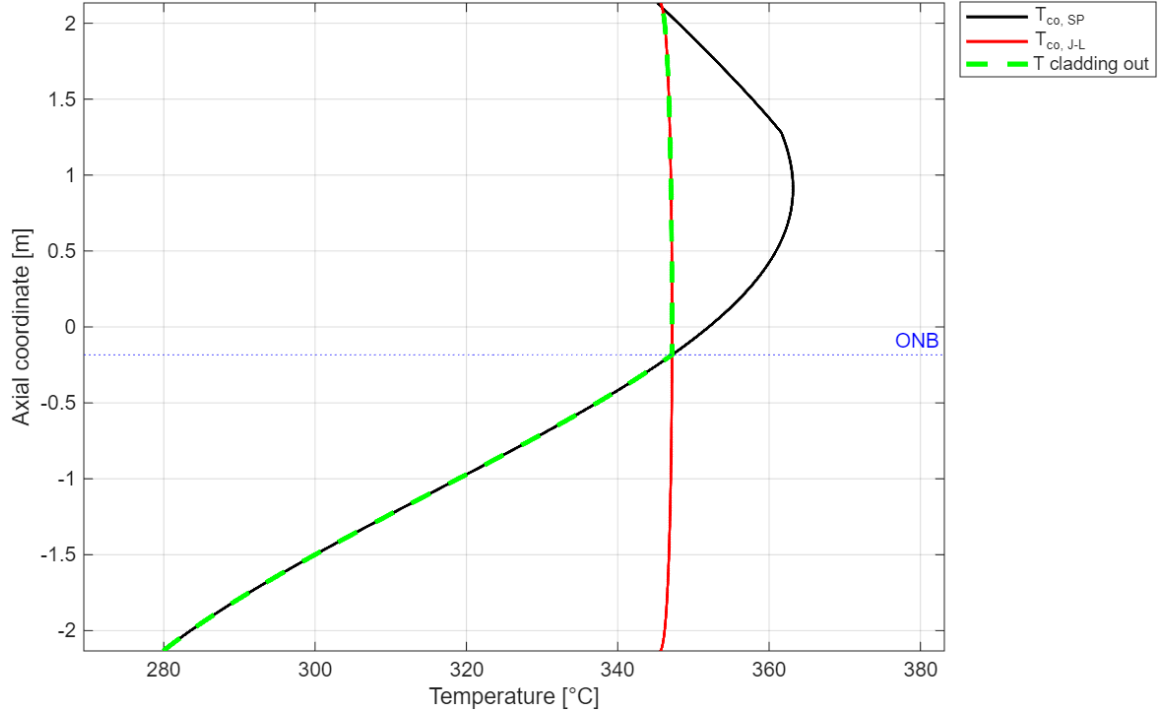


Figure 2: Cladding outer temperature evaluation and onset of nucleate boiling.

The onset of nucleate boiling is reached at:

$$z_{ONB} = -0.184 \text{ m} \quad (7.1)$$

After this point, the wall temperature is controlled by the nucleate boiling correlation. The ONB location is below the core mid-plane because the wall superheat required for nucleation is reached while the heat flux is already high and the bulk coolant is still subcooled. Once nucleate boiling starts, the wall heat transfer becomes more effective, which explains the nearly capped behaviour of the cladding outer temperature in the upper part of the core.

7.3 Quality Profiles

Figure 3 shows the equilibrium quality and the actual flow quality along the hot subchannel.

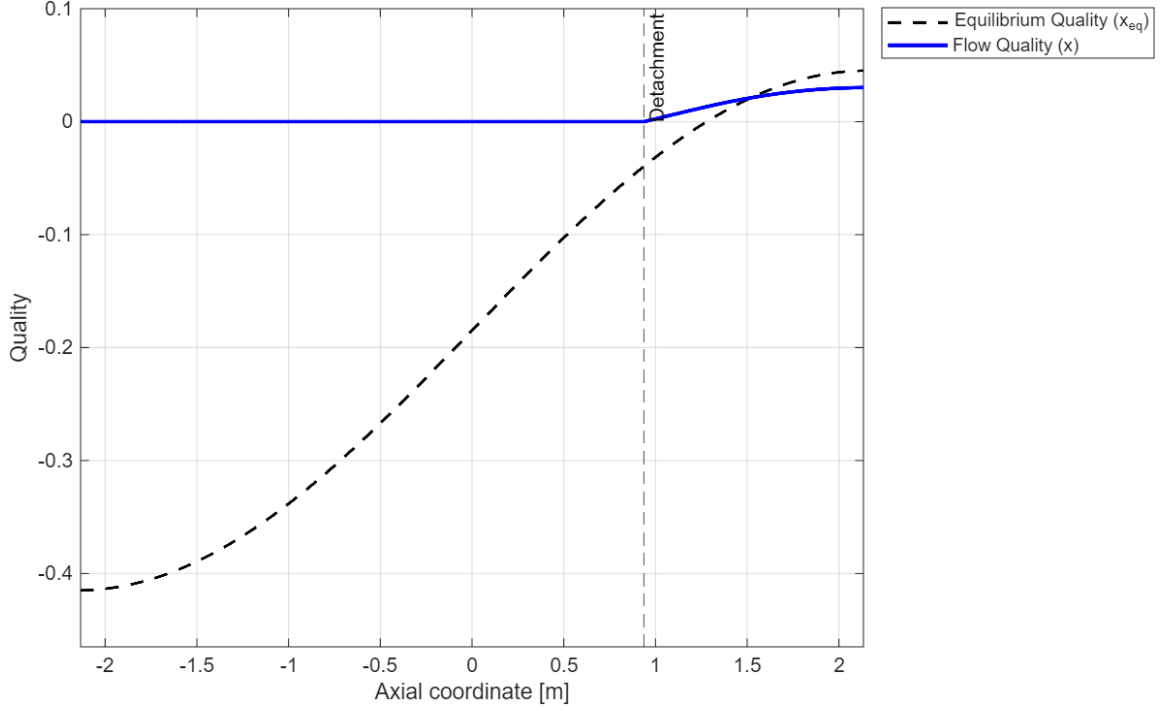


Figure 3: Equilibrium quality and actual flow quality along the hot subchannel.

Although nucleate boiling starts at $z_{ONB} = -0.184$ m, the actual flow quality remains zero until bubble detachment at $z_D = 0.937$ m. The maximum flow quality is $x_{max} = 0.03015$ at the outlet.

The difference between equilibrium quality and actual flow quality is a direct consequence of subcooled boiling. The equilibrium quality is based only on the local enthalpy balance, whereas the actual flow quality accounts for delayed vapour transport. In the wall-boiling region, part of the heat flux is still used for sensible heating of the liquid and part of the generated vapour condenses before becoming entrained in the bulk flow. For this reason, the actual flow quality remains lower than the equilibrium quality.

7.4 Void Fraction

Figure 4 compares the void fraction predicted by the Zuber–Findlay drift-flux model and by the Fauske slip-ratio model.

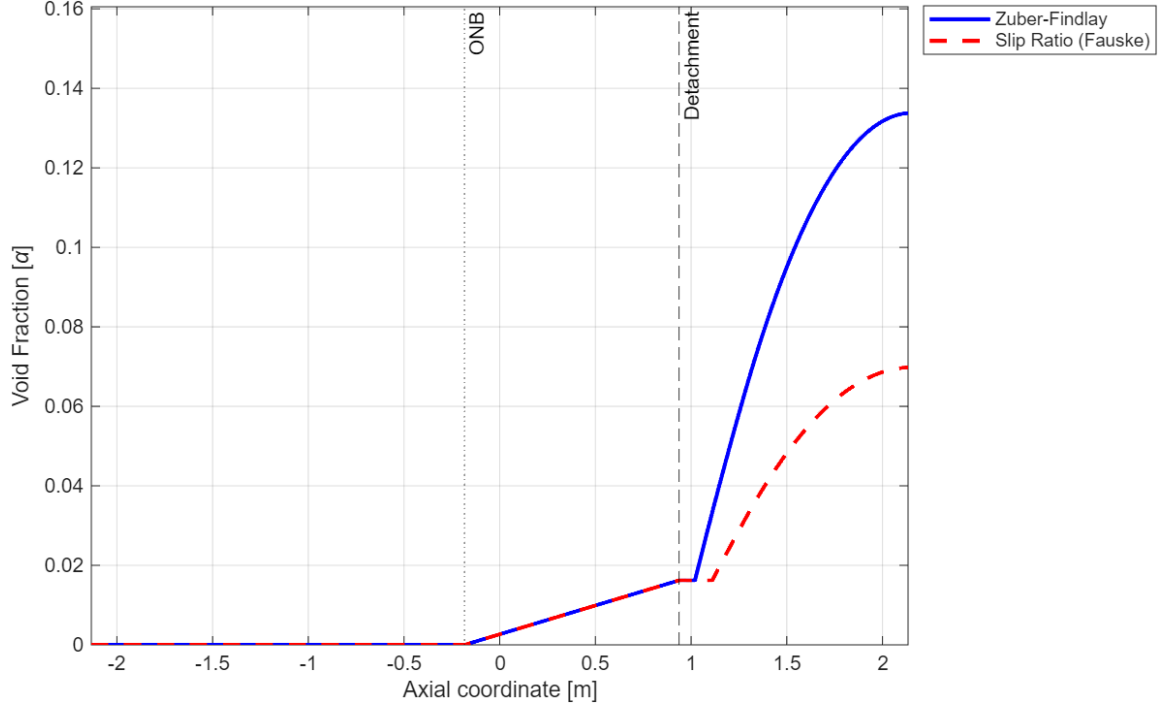


Figure 4: Void fraction profiles calculated with the Zuber–Findlay and slip-ratio models.

The maximum values are reached at the core outlet:

$$\alpha_{ZF,max} = 0.13377 \quad (7.2)$$

$$\alpha_{Slip,max} = 0.06978 \quad (7.3)$$

The void fraction increases much more rapidly than the actual flow quality because the vapour density is significantly lower than the liquid density. As a result, even a small vapour mass fraction occupies a non-negligible fraction of the subchannel volume.

The Zuber–Findlay model predicts a larger void fraction than the Fauske slip-ratio model. This is physically consistent with the different modelling assumptions: the drift-flux formulation accounts for phase distribution and drift velocity effects, whereas the slip-ratio model imposes a prescribed relative velocity between vapour and liquid. The comparison therefore provides a useful estimate of the modelling uncertainty associated with the void fraction evaluation.

7.5 Heat Partitioning Coefficient

Figure 5 shows the comparison between the constant Bowring coefficient and the Rouhani coefficient.

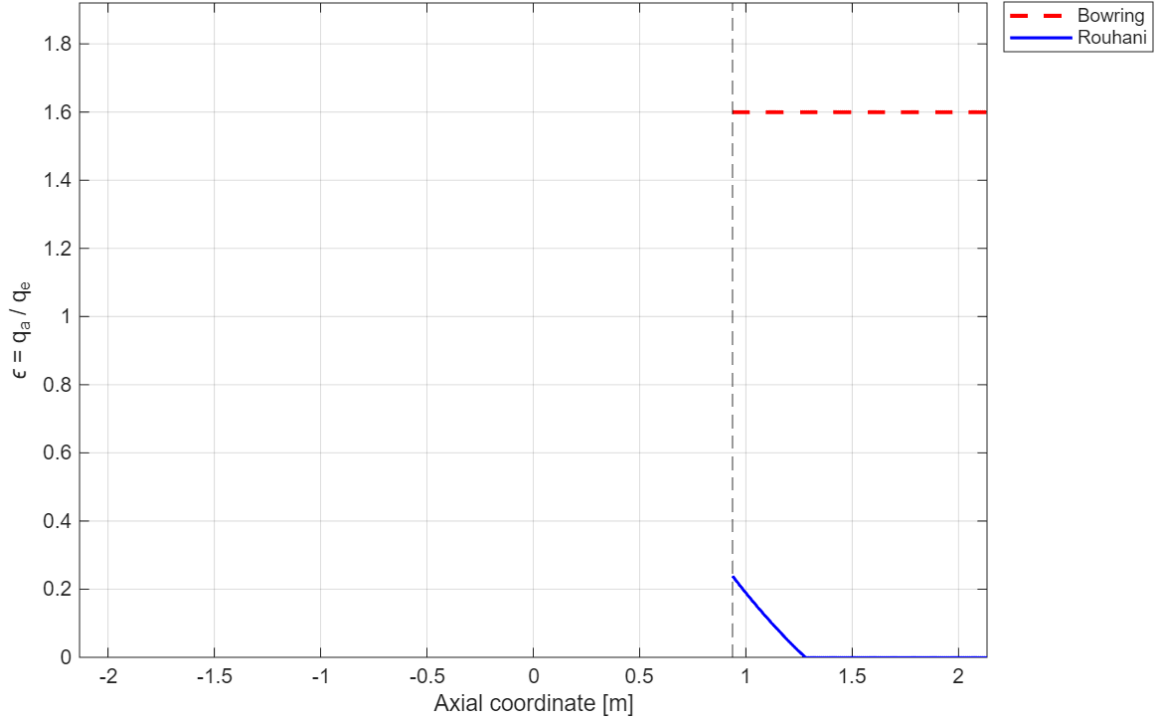


Figure 5: Comparison between the Bowring and Rouhani heat partitioning coefficients.

The Bowring model uses $\epsilon_{Bowring} = 1.6$ and is used in the final flow quality calculation. The Rouhani coefficient is calculated only for comparison and is limited to non-negative values.

This comparison is not used to replace the Bowring value in the final flow-quality calculation. Its purpose is to show the sensitivity of the heat-partitioning model to the local thermodynamic state. The decrease of the Rouhani coefficient reflects the progressive reduction of liquid sub-cooling as the coolant approaches saturation.

7.6 CHF and Actual Heat Flux

Figure 6 compares the actual surface heat flux with the corrected CHF curves. The plot starts from the ONB location.

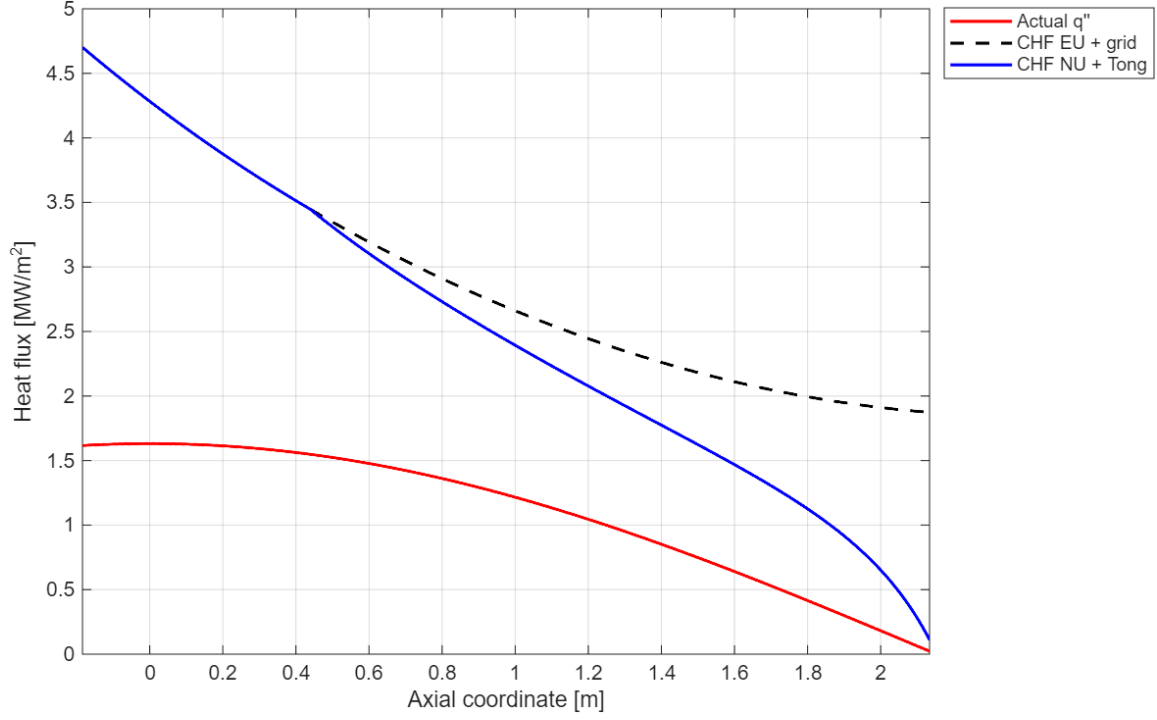


Figure 6: Actual surface heat flux and corrected CHF profiles from the ONB location.

The actual heat flux remains below the corrected critical heat flux along the whole boiling region considered in the plot. The CHF decreases in the upper part of the channel because the coolant enthalpy and equilibrium quality increase along the flow direction. The thermal margin is therefore not controlled only by the local imposed heat flux, but also by the progressively worsening coolant thermodynamic state.

7.7 DNBR Profile

The DNBR profile is shown in Figure 7. The plot includes both the nominal full-power reference limit and the overpower reference limit.

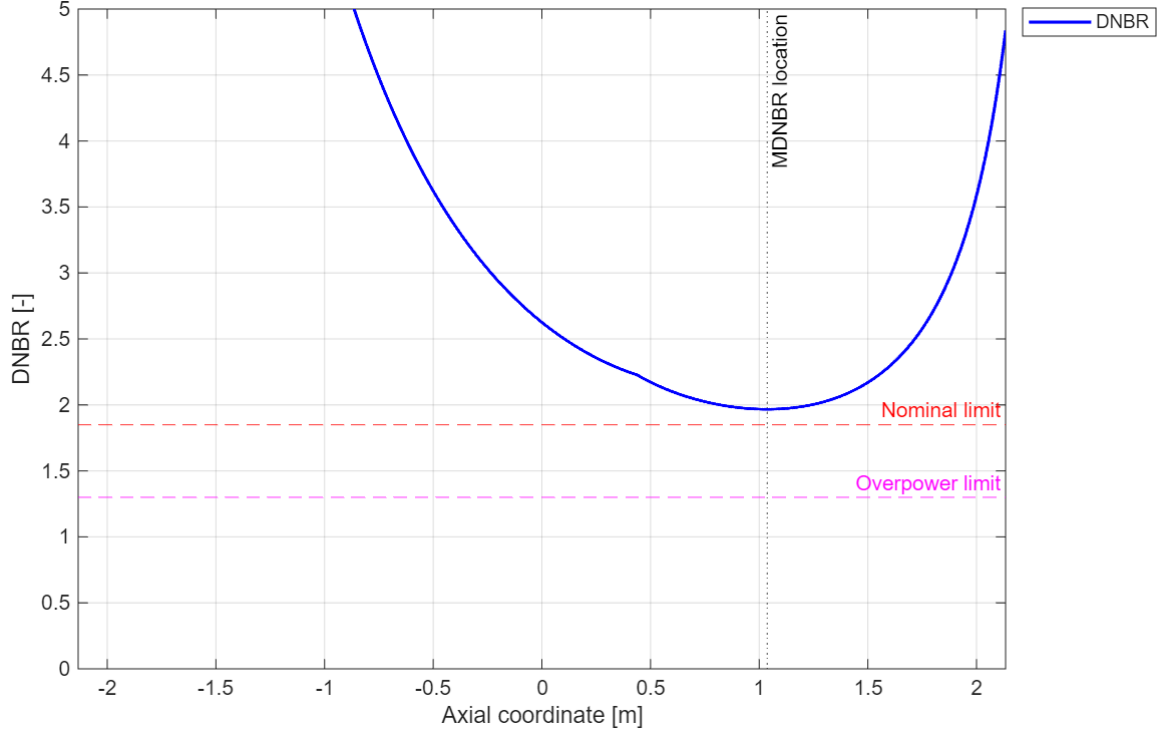


Figure 7: DNBR profile along the AP1000 hot subchannel.

The minimum DNBR obtained in the final calculation is $MDNBR = 1.967$, located at $z = 1.037$ m. It is higher than both reference limits, 1.85 and 1.30.

The MDNBR location is downstream of the axial power peak. This is expected because the DNBR is governed by the ratio between the corrected CHF and the actual heat flux. After the core mid-plane, the actual heat flux decreases, but the critical heat flux also decreases because the coolant is hotter and closer to saturated two-phase conditions. The minimum therefore results from the competition between the decreasing imposed heat flux and the decreasing CHF margin.

7.8 Heat Flux and Single-Phase Heat Transfer Coefficient

Figure 8 shows the axial surface heat flux and the single-phase heat transfer coefficient.

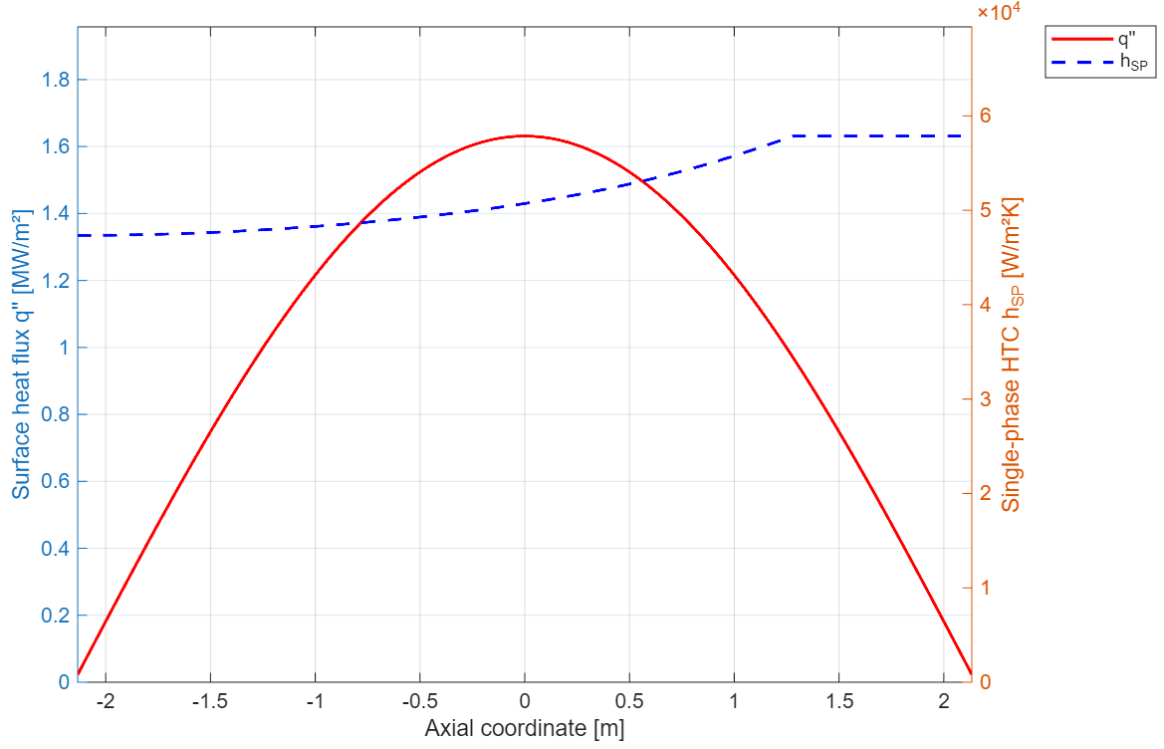


Figure 8: Surface heat flux and single-phase heat transfer coefficient along the hot subchannel.

The surface heat flux follows the chopped cosine axial profile and reaches its maximum close to the core mid-plane: $q''_{max} = 1.631 \text{ MW/m}^2$. The single-phase heat transfer coefficient changes more gradually because it depends on coolant properties, which evolve with the cumulative enthalpy rise rather than with the instantaneous local heat generation only. This coefficient is used consistently in the wall-temperature calculation, the Griffith detachment criterion and the heat-partitioning model.

7.9 Gap Thickness and Gap Conductance

Figure 9 shows the axial evolution of the hot fuel-cladding gap thickness and of the corresponding total gap conductance.

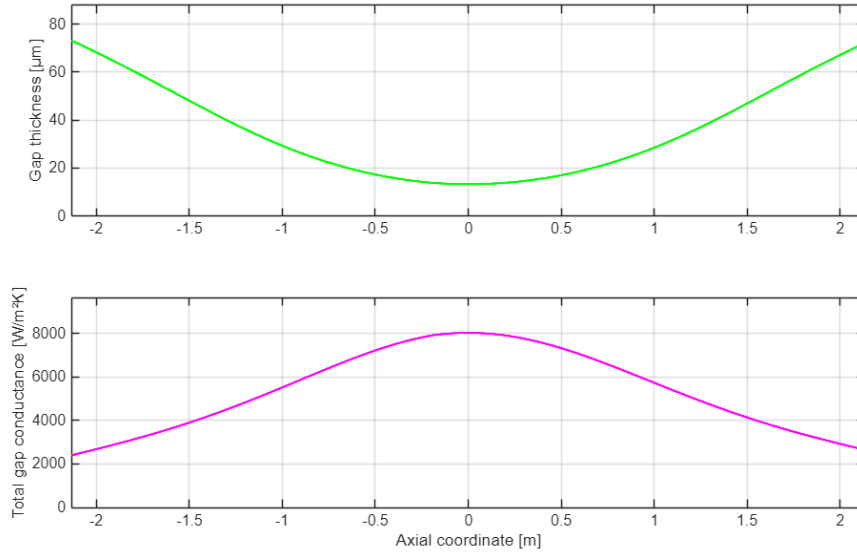


Figure 9: Hot gap thickness and total gap conductance along the fuel rod.

The hot gap thickness decreases in the high-power region because the fuel pellet expands thermally. As the gap becomes smaller, the conductive resistance across the helium gap is reduced and the effective gap conductance increases. This explains the opposite trends shown in Figure 9: the minimum gap thickness occurs close to the axial power peak, while the total gap conductance reaches its maximum in the same region.

This behaviour is consistent with the radial temperature profile. The central region of the core has the highest power density, the largest fuel expansion and the strongest fuel-to-cladding heat transfer. Moving toward the inlet and outlet, the local heat generation decreases, the fuel expansion becomes smaller, the gap opens again and the total gap conductance decreases.

Conclusions

A steady-state thermal-hydraulic analysis of the AP1000 hot subchannel under nominal Beginning of Life conditions was performed. The calculation included the axial coolant enthalpy rise, the cladding and fuel temperature profiles, the onset of nucleate boiling, bubble detachment, flow quality, void fraction, gap conductance, critical heat flux and DNBR.

The coolant reaches saturation in the upper part of the core. The outlet equilibrium quality is $x_{eq,out} = 0.04512$. The onset of nucleate boiling is predicted at $z_{ONB} = -0.184$ m, while bubble detachment occurs at $z_D = 0.937$ m. The actual flow quality remains zero up to the detachment point and then increases up to $x_{max} = 0.03015$ at the core outlet.

The maximum void fraction is $\alpha_{ZF,max} = 0.13377$ with the Zuber–Findlay drift-flux model and $\alpha_{Slip,max} = 0.06978$ with the Fauske slip-ratio model.

The maximum fuel centerline temperature is $T_{fcl,max} = 2094.11$ °C, below the reference uranium dioxide melting temperature $T_{melt,UO_2} = 2804.44$ °C. Therefore, the fuel temperature limit is satisfied.

The CHF was evaluated using the W-3 correlation, including the spacer grid correction, the 17×17 bundle correction factor and the Tong/Lin correction for non-uniform axial heat flux. The equivalent heated diameter used in the W-3 correlation is slightly above the nominal validity range of the correlation. Therefore, the calculated CHF and DNBR should be interpreted as empirical hot-subchannel estimates rather than as exact local boiling-crisis predictions.

The minimum DNBR obtained in the final calculation is $MDNBR = 1.967$ at $z_{MDNBR} = 1.037$ m. This value is higher than both the nominal full-power reference limit and the overpower reference limit:

$$1.967 > 1.85 \tag{7.4}$$

$$1.967 > 1.30 \tag{7.5}$$

Overall, all relevant thermal limits are respected. The AP1000 hot subchannel therefore operates with a positive thermal margin under the nominal conditions analysed in this work.

The model remains conservative because cross-flow and lateral turbulent mixing are neglected. At the same time, the results should be interpreted as a one-dimensional hot-subchannel estimate: multidimensional effects, spacer-induced mixing and detailed subchannel redistribution are not explicitly resolved.

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